



UNIVERSITI MALAYA

WIF2003 WEB PROGRAMMING SEMESTER 2, 2025/2026 PROJECT REPORT

Web: **Entertainment Recommendation System**

Occurrence: **2**

Group: **Team 02**

Lecturer: **Dr. Uzair Iqbal**

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TABLE OF CONTENTS

1.1 Project Description and Objective.....	5
1.2 Project Team.....	6
1.3 Main Deliverables.....	6
1.4 List of Features.....	7
1.5 Quality Requirements.....	8
1.6 Systems Constraints/Assumptions.....	9
2.1 Approaches and Techniques Used to Design the Front-End	10
2.2 Graphical User Interface Design.....	15

LIST OF FIGURES

Figure 1: Register page.....	15
Figure 2: Login page.....	15
Figure 3: Profile page.....	16
Figure 4: Edit profile.....	16
Figure 5: Delete account.....	17
Figure 6: Trending page.....	17
Figure 7: Explore page.....	18
Figure 8: Search movie by title.....	18
Figure 9: Movie description.....	19
Figure 10: Posting review.....	19
Figure 11: Adding to watchlist.....	20
Figure 12: Watchlist page.....	20
Appendix A: ERD Diagram.....	22
Appendix B: Sitemap.....	22

LIST OF TABLES

Table A: Project Team.....	6
Table B: Main Deliverables.....	6

SECTION 1.0 INTRODUCTION

1.1 Project Description and Objective

PopcornPicks is an Entertainment Recommendation System that is catered for users who would like to browse movies and series. The system recommends the Top 10 movies globally, and allows users to search for movies and filter based on genre.

The objectives of PopcornPicks are as follows:

1. To provide real-time movie recommendations by integrating external data sources using OMDB API.
2. To allow user to view movies' descriptions and current ratings.
3. To allow user to search movies and series based on title and filters.
4. To allow user to add and remove movies from watchlist.
5. To facilitate user engagement through a local persistence-based rating and review system.

1.2 Project Team

Table A: Project Team

Name	Matric No	Module	Features
Khalisya	23001372	General	Account registration, log in, log out and manage sessions
Nadya	22095142	Profile Management	View profile, update profile, delete profile and change password
Dawna	23002262	Entertainment Chart	Integrate OMDB API, display Top 10 Movies
Nurdeena	23002721	Search Recommendation	Search query, filter by genre query
Sakinah	23002367	Watchlist	View, add and remove watchlist
Hasna	22107029	User rating	Display and add user ratings

1.3 Main Deliverables

Table B: Main Deliverables

No.	Main Deliverable	Links
1.	Project Report	https://github.com/dwnsky/popcornpicks/report1.pdf
2.	Source Code	https://github.com/dwnsky/popcornpicks
3.	ERD Diagram	https://github.com/dwnsky/popcornpicks/blob/main/erd.png
4.	Database Sample	https://github.com/dwnsky/popcornpicks/blob/main/

		n/database_export.json
5.	Sitemap	https://github.com/dwnsky/popcornpicks/blob/main/sitemap.png
6.	Unit Testing	https://github.com/dwnsky/popcornpicks/blob/main/Unit%20Testing.pdf

1.4 List of Features

1.4.1 Top Movie Titles

The system features a Trending landing page that automatically displays a curated list of top-rated or popular movies. This ensures users are immediately presented with high-quality content recommendations upon entering the application.

1.4.2 Search Movie by Title

Users can look up specific films using the search bar on the Explore page. The system performs asynchronous API calls to the OMDB database to fetch and display real-time results, including titles, years, and posters matching the user's query.

1.4.3 Filter by Genre

To enhance discovery, the Explore page includes genre-specific "pills" (e.g., Action, Comedy, Horror). Clicking a pill filters the search results or trending list, allowing users to narrow down movies based on their preferred entertainment category.

1.4.4 View Movie Details

By clicking on a movie card, users are directed to a dedicated description page (moviedesc.html). This feature provides in-depth information, including a detailed storyline, IMDB ratings, and a direct link to watch the movie trailer.

1.4.5 Ratings

On the movie description page, authenticated users can interact with the content by providing a star rating and writing personal reviews. This feedback is captured and displayed to help build a community-driven recommendation environment.

1.4.6 Watchlist

Users can personalize their experience by adding movies to a "Watchlist." This feature allows for the saving of titles for future viewing. Users can view their full collection on a dedicated Watchlist page and remove titles once they have been watched.

1.5 Quality Requirements

1.5.1 Functional Suitability

The system must provide functions that meet the stated objectives of movie discovery which includes accurate retrieval of movie data through the OMDB API and the capability of users to manage a personal watchlist and submit reviews.

1.5.2 Performance Efficiency

The web must demonstrate “fast loading and response times” which is achieved by using asynchronous Javascript to ensure that API data fetching does not block the user interface.

1.5.3 Security

User data must be protected to ensure enhanced security. In Phase 1, this involves session management to restrict profile access. In Phase 2, this will extend to protecting the MongoDB database and user credentials.

1.5.4 Interaction Capability

The system must be “well-organized and easy to navigate” where users should clearly understand their current location within the web app through visual cues like “active” nav-pills and consistent Bootstrap design elements.

1.6 Systems Constraints/Assumptions

1.6.1 Assumptions

- Users have basic familiarity with web navigation and standard UI patterns, such as using a search bar or clicking icons for actions.
- The data provided by the OMDB API (ratings, plots, posters) is accurate and up-to-date.
- Users will access the site on a screen with at least a 320px width which is mobile-responsive minimum and Bootstrap grid will handle resizing across standard devices.

1.6.2 Constraints

- The system requires a stable internet connection to function, as movie posters and metadata are fetched in real-time from external servers.
- The application does not host video files directly but relies on external links for trailers. This is a constraint because the application cannot control if a trailer is taken down or restricted in certain regions.
- The application relies on the OMDB API, which has a daily request limit that limits the volume of traffic the system can handle during testing and presentation.

SECTION 2.0 FRONT-END

2.1 Approaches and Techniques Used to Design the Front-End

2.1.1 Front-End Development Approach

The front-end of PopcornPicks was developed using a combination of HTML, CSS, JavaScript, and Bootstrap to create an interactive and responsive web application [3]. The development approach focused on building a user-friendly interface with consistent navigation, organized layouts, and responsive design to improve the ease of use and overall comfort of users while interacting with the system. The interface was designed to ensure that users can navigate between pages smoothly and access content efficiently.

In addition, the user interface (UI) was designed using a minimalist and modern approach. Unnecessary visual clutter and excessive design elements were avoided in order to create a clean and organized appearance. A consistent color scheme, readable typography, adequate spacing, and clear arrangement of components were applied to enhance overall user experience. This design approach helps users focus on the main content and functionalities of the system.

2.1.2 HTML for Structure and Content

HTML was used as the foundation for structuring and organizing the content of the web application [4]. Different HTML elements were utilized to divide the website into logical sections such as headers, navigation bars, content containers, forms, buttons, and interactive sections. This structured organization helped create a clear layout and improved the maintainability of the front-end development process.

Semantic HTML elements were also applied to improve readability and organization of the code. Elements such as `<header>`, `<section>`, `<div>`, `<form>`, and `<button>` were used to separate different parts of the interface and ensure that the content hierarchy remained clear and consistent throughout the system. Navigation links were implemented using anchor (`<a>`) elements to allow smooth movement between pages within the application.

Furthermore, IDs and classes were added to elements to assist in styling with CSS and handling dynamic interactions with JavaScript. These identifiers enabled for integration of interactive functionality, dynamic content rendering and responsive behaviour into the system.

2.1.3 CSS for Styling and User Interface Design

CSS was used throughout the PopcornPicks system to establish a consistent visual identity and enhance the overall aesthetic of the interface. A dark-themed color scheme was applied across all pages, with a two-tone background using a CSS linear gradient to create a seamless, immersive feel as users scroll. The primary accent color, #612D53 (a deep plum-purple) was used consistently across interactive elements such as buttons, active navigation states, focus rings, and profile borders to maintain visual coherence.

Custom CSS classes were written to style specific components across the system, including authentication forms, navigation bars, profile sections, movie cards, genre filter buttons, and action buttons. Each component was styled with matching background colors, border styles, and rounded corners to ensure a unified appearance throughout the application. Hover and focus states were also defined for interactive elements such as login buttons and form inputs to provide visual feedback during user interaction, improving overall usability.

Flexbox was used extensively to manage the alignment and positioning of elements within the interface, including centering page content vertically and horizontally, aligning header navigation items, and arranging the search bar and genre pill buttons. For the entertainment listing and watchlist pages, CSS Grid was applied using `repeat(auto-fill, minmax(200px, 1fr))` to create a responsive multi-column movie card layout that automatically adjusts based on available screen width, without relying on explicit media queries.

The header was styled with a fixed position to remain visible during scrolling, ensuring users always have access to navigation regardless of where they are on the page. Profile-related elements such as avatar images were styled using `border-radius` and `object-fit` properties to display circular, properly cropped images consistently across the header and profile page. A custom search bar was also styled using Flexbox with a pill-shaped border, transparent input background, and icon positioning to create a clean, integrated search experience consistent with the overall dark theme.

2.1.4 JavaScript for Interactivity

JavaScript was used throughout the PopcornPicks system to handle dynamic behaviour and client-side logic across all major features of the application. While HTML and CSS defined the structure and appearance of the interface, JavaScript was responsible for responding to user actions, processing data, and updating page content dynamically without requiring full page reloads.

Client-side form validation was implemented on the registration and login pages to check user inputs before any data was processed. Validation logic covered cases such as empty fields, mismatched passwords, minimum password length, and invalid email format. When a check failed, error messages were displayed

dynamically by manipulating the DOM to show or hide alert elements, providing users with immediate feedback without leaving the page.

User authentication and session management were handled by communicating with the back-end database, where user credentials and session states are stored and verified server-side. Functions were written to manage registration, login, logout, and session checking, each sending and retrieving data from the database as needed. A session check function was also implemented to redirect unauthenticated users away from protected pages. Profile management features including updating the display name, uploading a profile photo, changing the password, and deleting the account were handled through JavaScript by sending update requests to the database and reflecting the changes across the interface in real time.

For the entertainment features, JavaScript was used to make asynchronous API calls to the OMDB API using the `fetch()` method to retrieve real-time movie data. The fetched data was then dynamically rendered into the DOM as movie cards, allowing the trending movies, search results, and genre-filtered content to update without refreshing the page. Event listeners were attached to the search input and genre filter buttons to trigger the appropriate fetch calls in response to user interaction. The watchlist feature was managed by storing and retrieving each user's saved movies from the database, with functions written to add, display, and remove items from a user's personal list. Additionally, a star-based rating and review system was implemented using dynamically generated DOM elements, allowing users to submit and view ratings and written reviews for individual movies, with all data stored persistently in the database.

2.1.5 Bootstrap Framework

PopcornPick uses Bootstrap as an open-source front-end framework that provides pre-styled components such as buttons, form, and navigation [5]. By linking Bootstrap to HTML, the pre-styled components allow rapid building the website design by reducing time on CSS development and without requiring additional configuration.

A wide range of Bootstrap's utility classes were used to control the spacing, alignment, text style and visibility of the elements. These classes included *d-flex*, *align-items-center*, *justify-content-center*, *text-center* and *mb-3* to decrease the amount of repetitive custom CSS and to keep the layout structure uniform. The form control and button classes provided by Bootstrap were used for form elements and buttons on the authentication screens, profile and movie description screens, and custom overrides were added in *app.css* to reflect the dark theme and accent color of the system.

Additionally, Bootstrap Icons was also added as its own icon library to get scaled icons across the interface. Navigation actions, like trending, explore, watchlist, login and logout, were depicted using icons. These icons enhanced the visual of the interface and enabled users to instantly understand the function of each component without needing to depend purely on the textual labels.

2.1.6 Navigation and User Experience Design

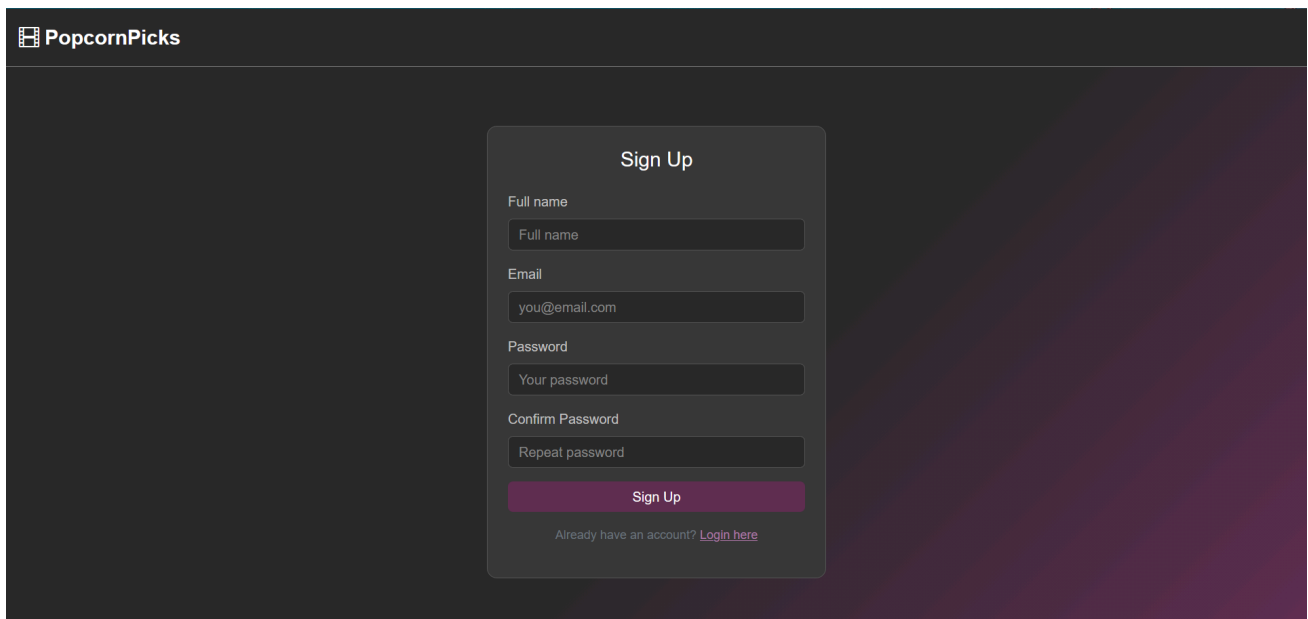
User experience design is an aspect of creating products that provide meaningful and valuable experiences to users. Several aspects like visual design, navigation, and interaction quality have an impact on user experience to increase user satisfaction [6]. Therefore, good navigation must be achieved to have good user experience.

PopcornPicks' navigation structure was designed to give users a clear and consistent way to access all key areas of the app. A fixed header at the top of each page with the brand logo, navigation pill links for 'Trending', 'Explore' and 'Watchlist' and user action area on the right. The current page is highlighted to the user using system accent color to always indicate the current page. The user action area dynamically shows either a log in button or a user profile avatar, giving users a quick visual indication of their log in status.

The overall user experience has been created with simplicity as a priority, ensuring that the pages remain clear, uncluttered and the content is centered and spaced. Interactive elements, like buttons and form inputs, have hover and focus states to give visual feedback when interacting with them. Errors and success messages appear directly on the page, rather than to a new page, and do not interrupt the flow of the application, providing a smooth and comfortable experience.

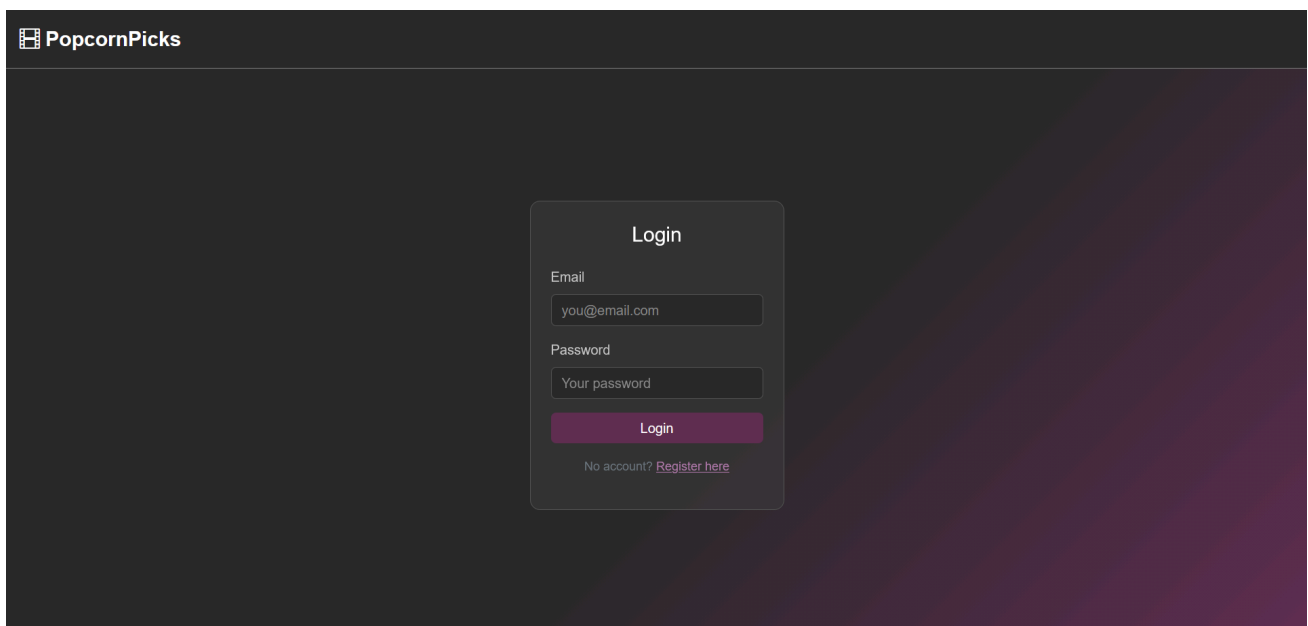
2.2 Graphical User Interface Design

Figure 1: Register page



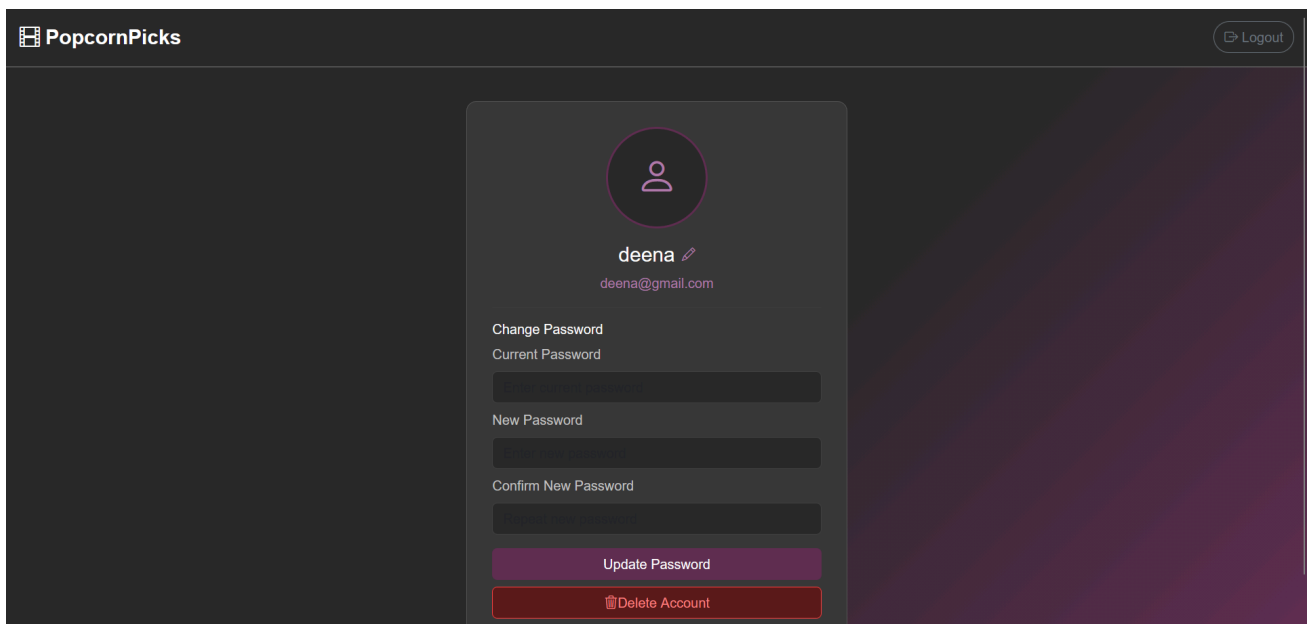
The image shows a web page for 'PopcornPicks' with a dark background. In the top left corner, there is a logo consisting of a film strip icon and the text 'PopcornPicks'. Centered on the page is a light gray rounded rectangle containing a 'Sign Up' form. The form has the title 'Sign Up' at the top. Below the title are four input fields: 'Full name' with the placeholder 'Full name', 'Email' with the placeholder 'you@email.com', 'Password' with the placeholder 'Your password', and 'Confirm Password' with the placeholder 'Repeat password'. Below these fields is a purple button labeled 'Sign Up'. At the bottom of the form, there is a link that says 'Already have an account? [Login here](#)'.

Figure 2: Login page

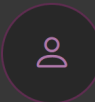


The image shows a web page for 'PopcornPicks' with a dark background. In the top left corner, there is a logo consisting of a film strip icon and the text 'PopcornPicks'. Centered on the page is a light gray rounded rectangle containing a 'Login' form. The form has the title 'Login' at the top. Below the title are two input fields: 'Email' with the placeholder 'you@email.com' and 'Password' with the placeholder 'Your password'. Below these fields is a purple button labeled 'Login'. At the bottom of the form, there is a link that says 'No account? [Register here](#)'.

Figure 3: Profile page



PopcornPicks Logout


deena ✎
deena@gmail.com

Change Password

Current Password

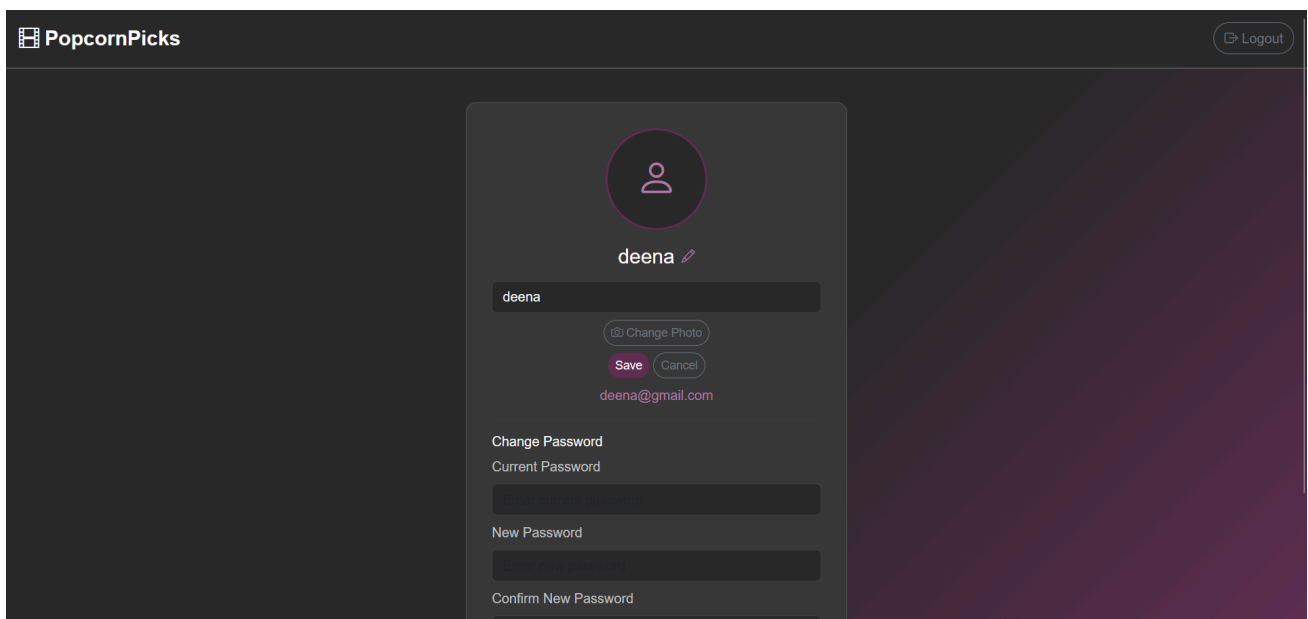
New Password

Confirm New Password

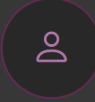
Update Password

Delete Account

Figure 4: Edit profile



PopcornPicks Logout


deena ✎
deena@gmail.com

deena

Change Photo

Save Cancel

Change Password

Current Password

New Password

Confirm New Password

Figure 5: Delete account

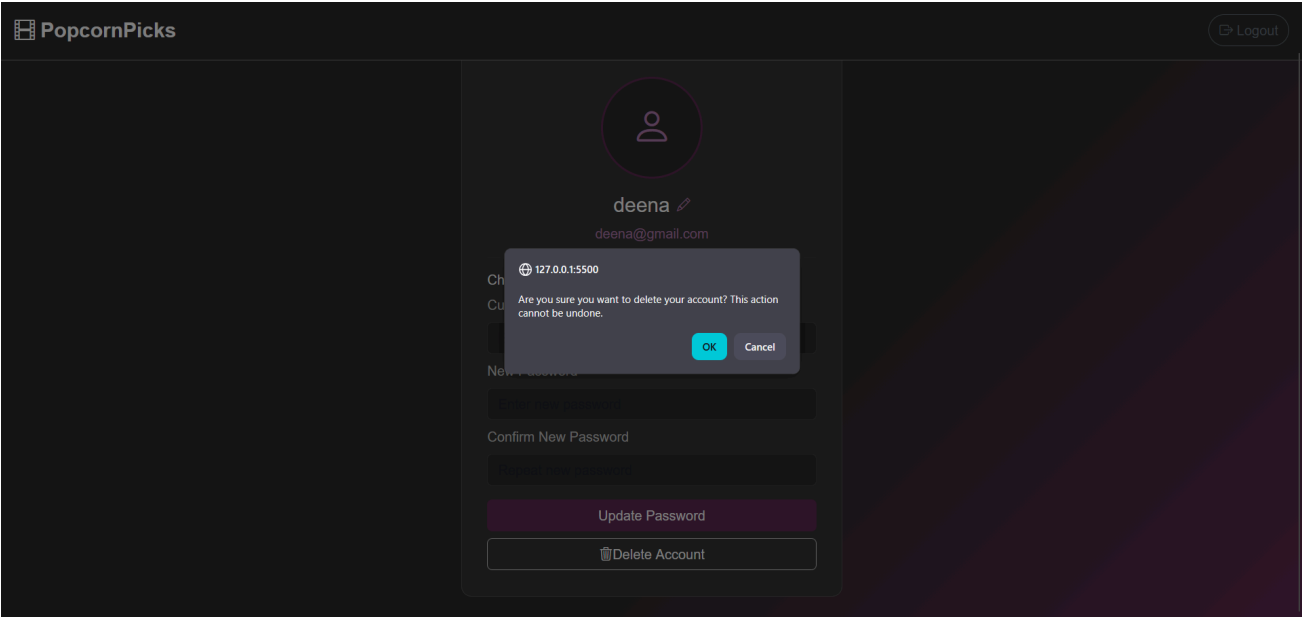


Figure 6: Trending page

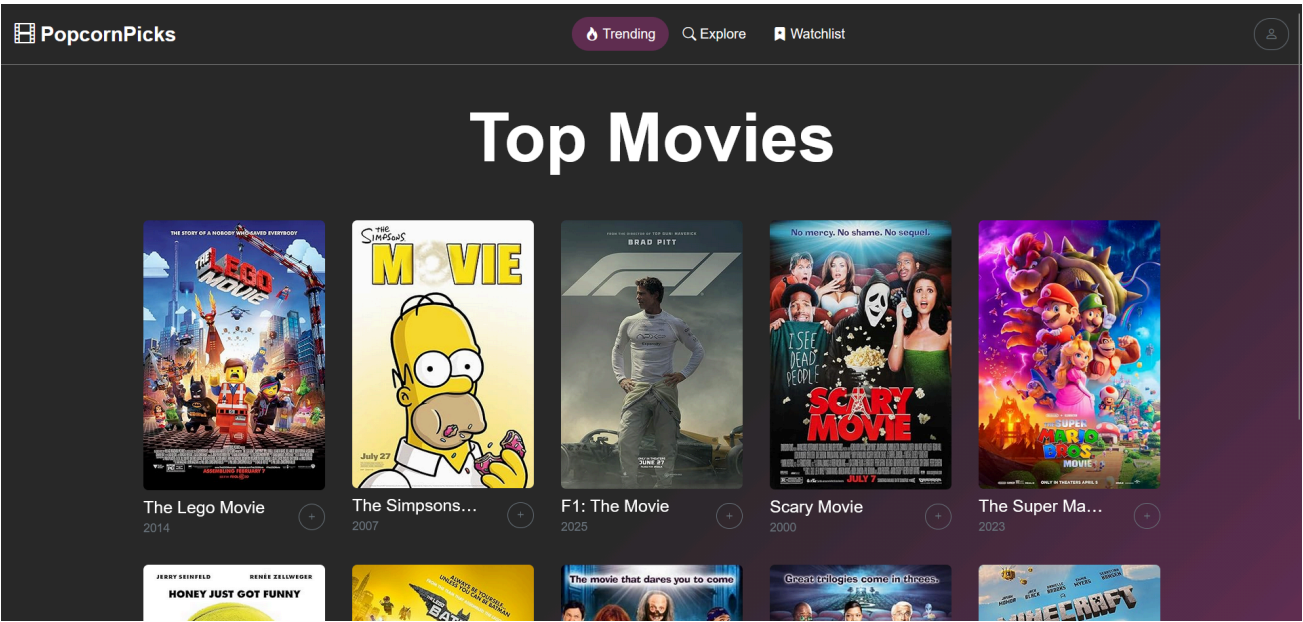


Figure 7: Explore page

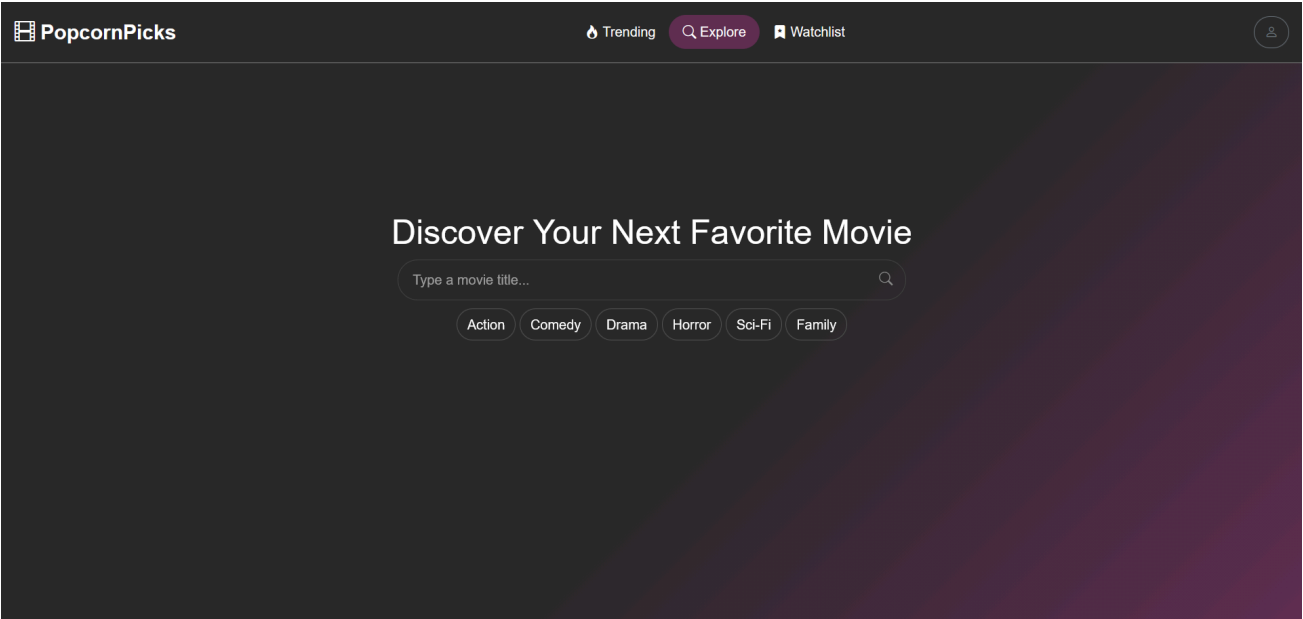


Figure 8: Search movie by title

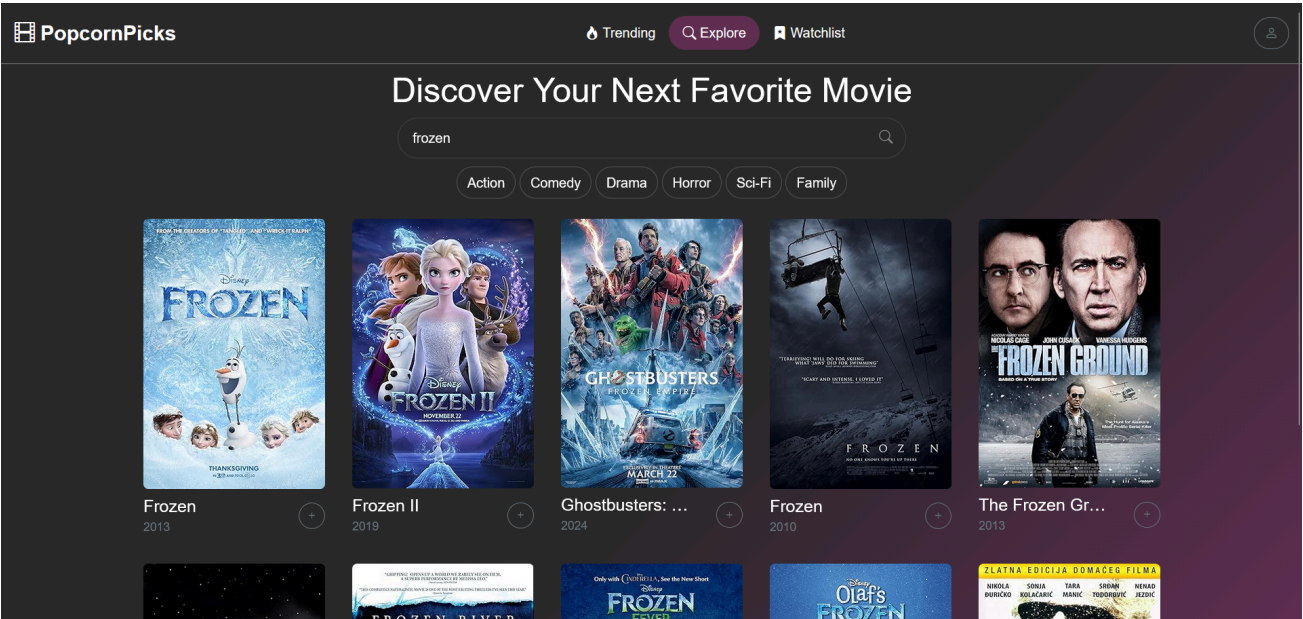


Figure 9: Movie description

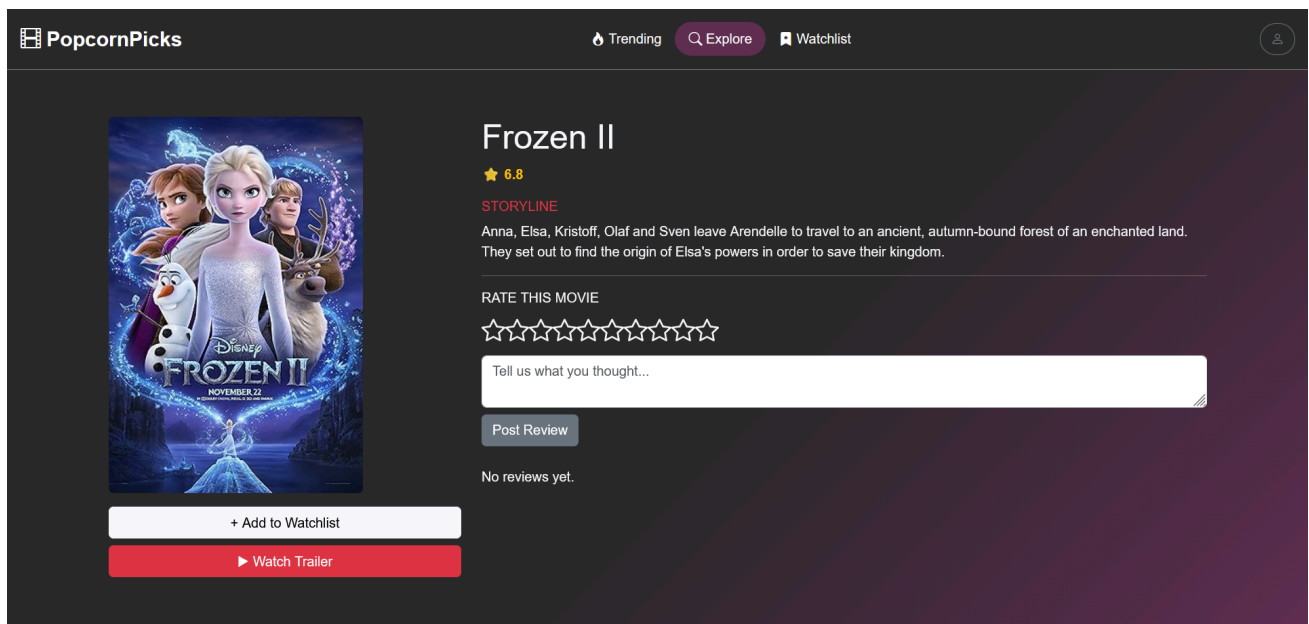


Figure 10: Posting review

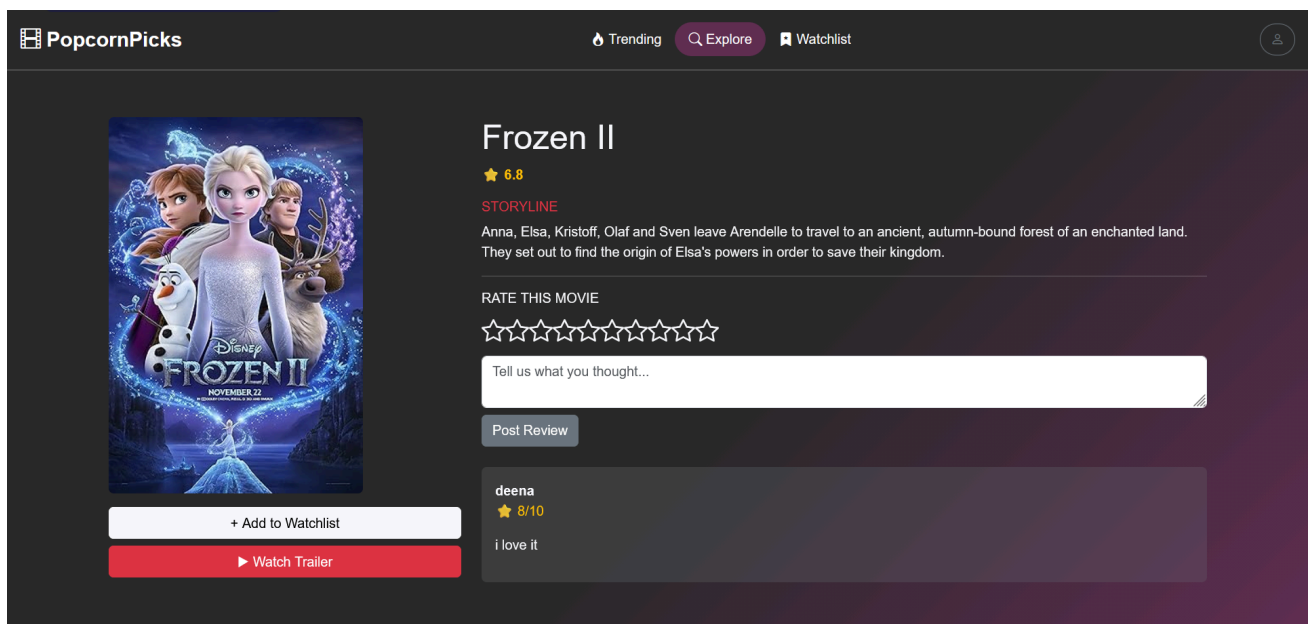


Figure 11: Adding to watchlist

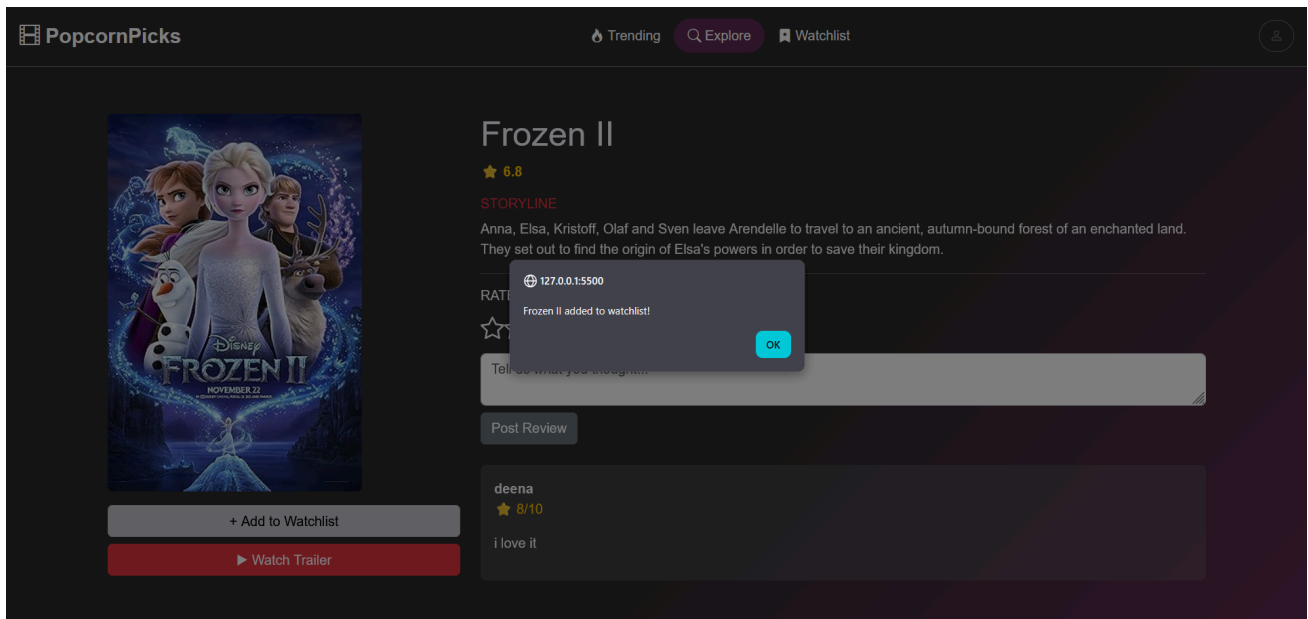
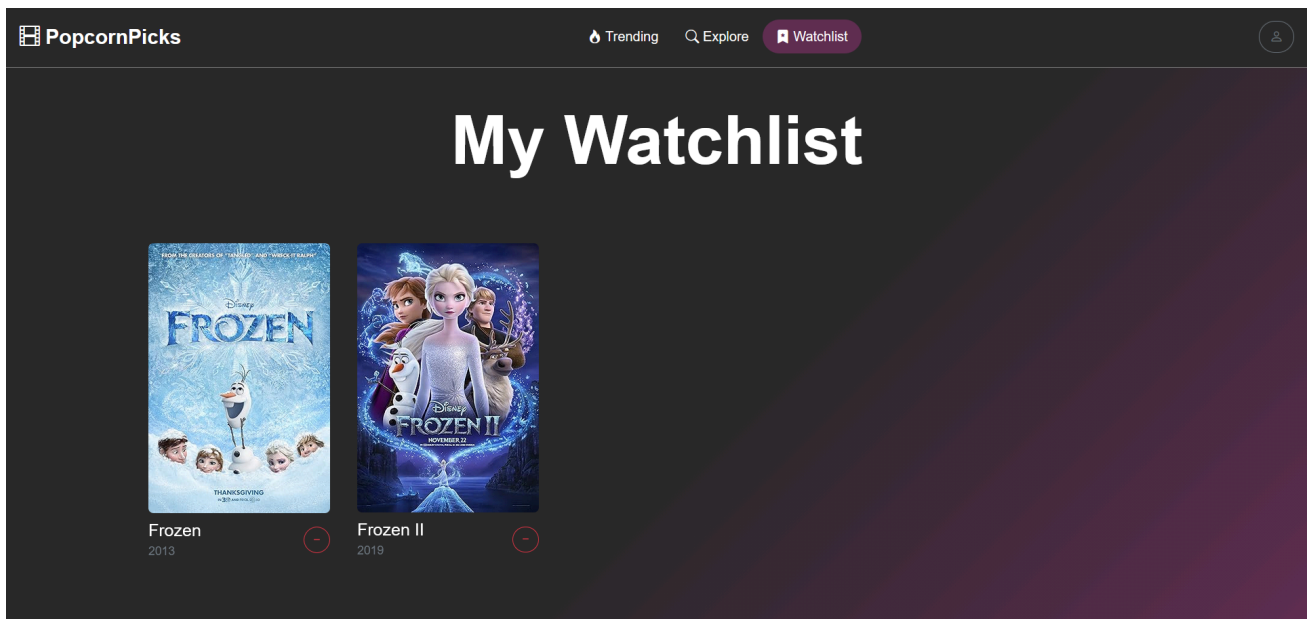


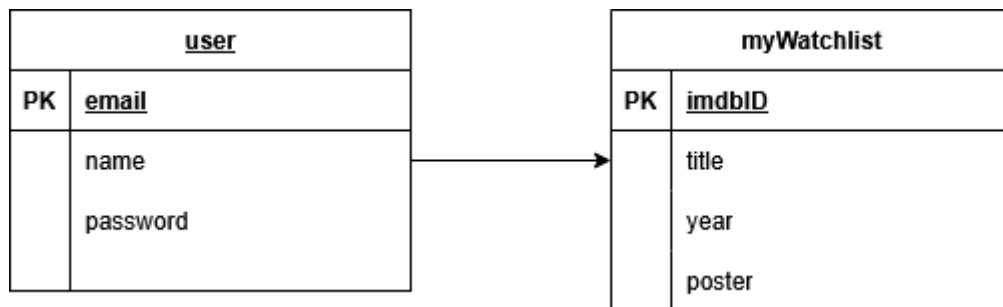
Figure 12: Watchlist page



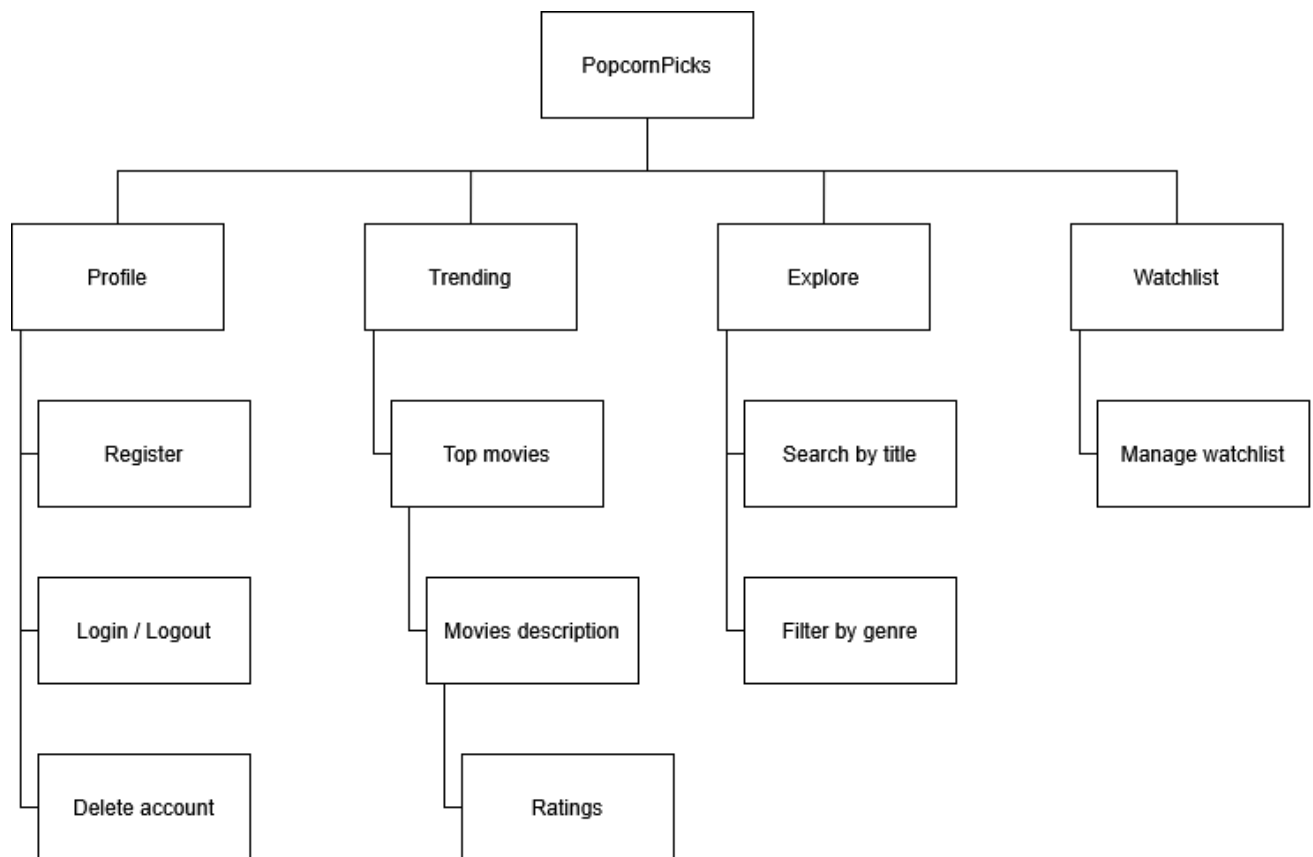
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APPENDIX



Appendix A: ERD Diagram



Appendix B: Sitemap